CSE 4074 Quiz

02. 04. 2021

1 Sessions (35 minutes)

Cheat Sheets are not allowed!

“On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues”

Take it easy!

Q1. (20 pts) For each of the following, indicate the class Θ(g(n)) the function belongs to. Use the simplest g(n) possible.

(a) $\sum_{i=1}^{3n} \log_3 i$ (b) $\ln(\pi^{en}) + \ln(n^{\pi e})$ (c) $10^{n/2} + n^3 3^n$ (d) π^e

Q2. (20 pts) For each of the following indicate whether it is true or false

(i) $\log_2 n^2+1 \in O(n)$.

(ii) $\sqrt{n(n+1)} \in \Omega(n)$.

(iii) $(2n)! \in \Theta(n!)$.

(iv) $\log_2 \sqrt{n}$ and $(\log_2 n)^2$ are of the same asymptotical order.

Q3. (30 pts) How many multiplications (in terms of n) will be performed in the following function? Write a recurrence relation and solve it by backward substitution.

```
function f(n)
  if n <= 1 return 2
  else    return    f(n-1)*f(n-1)*f(n-1)
```

Q4. (30 pts) What is the time complexity of the following function?

```
void func(int n) {
        int i = 1;
        int x = 0;
        int count = 0;

        while (i < n){
                x = x + 1;
                i = i*3;
        }
        for(int j = 1; j <= n; j++)
                for(int k = 1; k <= j; k++)
                        for(int m = 1; m <= x; m++)
                                count = count + 1;
}
```

Q1 (a) $\log_3 1 + \log_3 2 + \dots + \log_3 3n = \log_3 (3n)! \in \Theta(3n \log(3n))$
$= \Theta(n \log n)$

(b) $e^n \ln \pi + \pi^e \ln n \in \Theta(n)$

(c) $10^{n/2} + n^3 3^n$ $\lim_{n\to\infty} \frac{10^{n/2}}{n^3 3^{n/2}} = \infty$ Therefore it is in $\Theta(10^{n/2})$

(d) $\pi^e \in \Theta(1)$

Q2 (i) $2\log_2 n + 1 \in \Theta(n)$ True. $\log_2 n$ grows no faster than n.

(ii) True $\sqrt{n(n+1)}$ & n has the same asymptotic growth.

(iii) False $\lim \frac{(2n)!}{n!} = \infty$

(iv) False $\lim \frac{(\log_2 n)^n}{\frac{1}{2}\log_2 n} = \infty$

Q3 $M(1) = 0$

$M(n) = 3M(n-1) + 2$

$M(n) = 3(3M(n-2)+2) + 2 = 3^2 M(n-2) + 2\cdot 3 + 2$

$= 3^2(3M(n-3)+2) + 2\cdot 3 + 2 = 3^3 M(n-3) + 2\cdot 3^2 + 2\cdot 3 + 2$

$= 3^{n-1} \underbrace{M(n-(n-1))}_{0} + 2(3^{n-2} + 3^{n-3} + \dots 1)$

$= 2 \cdot \frac{3^{n-1} - 1}{3-1} = 3^{n-1} - 1$ //

Q4 $x = \log_3 n$

$T(n) = \sum_{j=1}^{n} \sum_{k=1}^{j} \sum_{m=1}^{x} 1 = \sum_{j=1}^{n} \sum_{k=1}^{j} x = \sum_{j=1}^{n} xj$

$= x(1+2+\dots n) = \frac{x\,n(n+1)}{2} = \frac{\log_3 n \cdot n(n+1)}{2} \in \Theta(n^2 \log n)$